

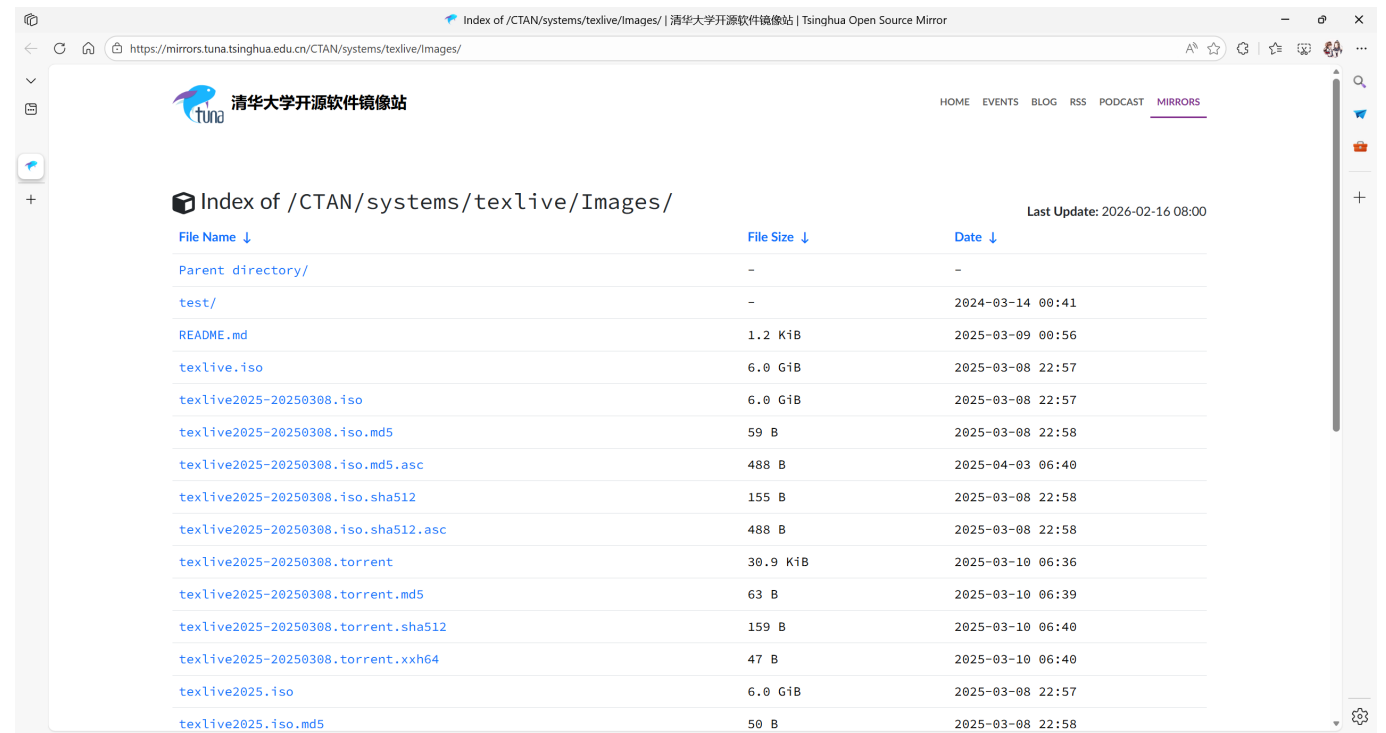
Latex+TexLive+VSCode环境配置

1. 安装TexLive

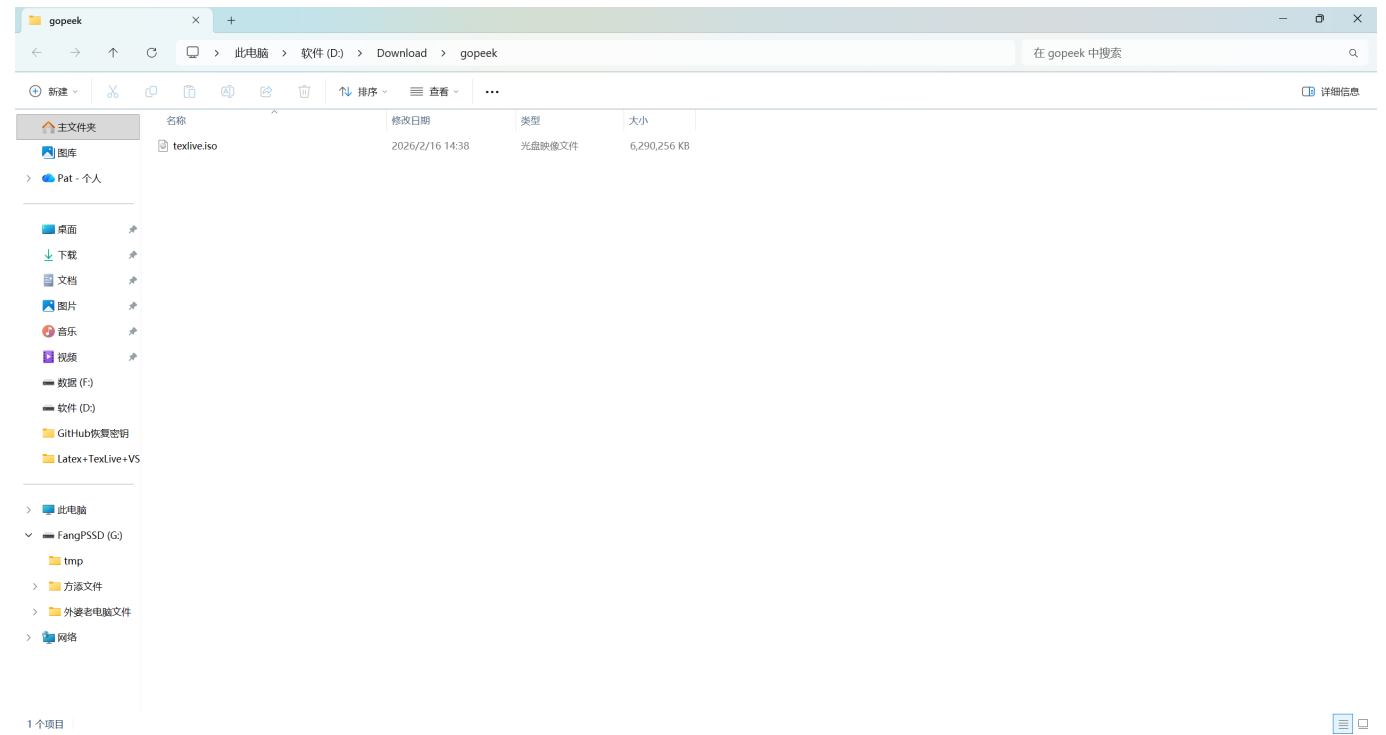
1.1 下载

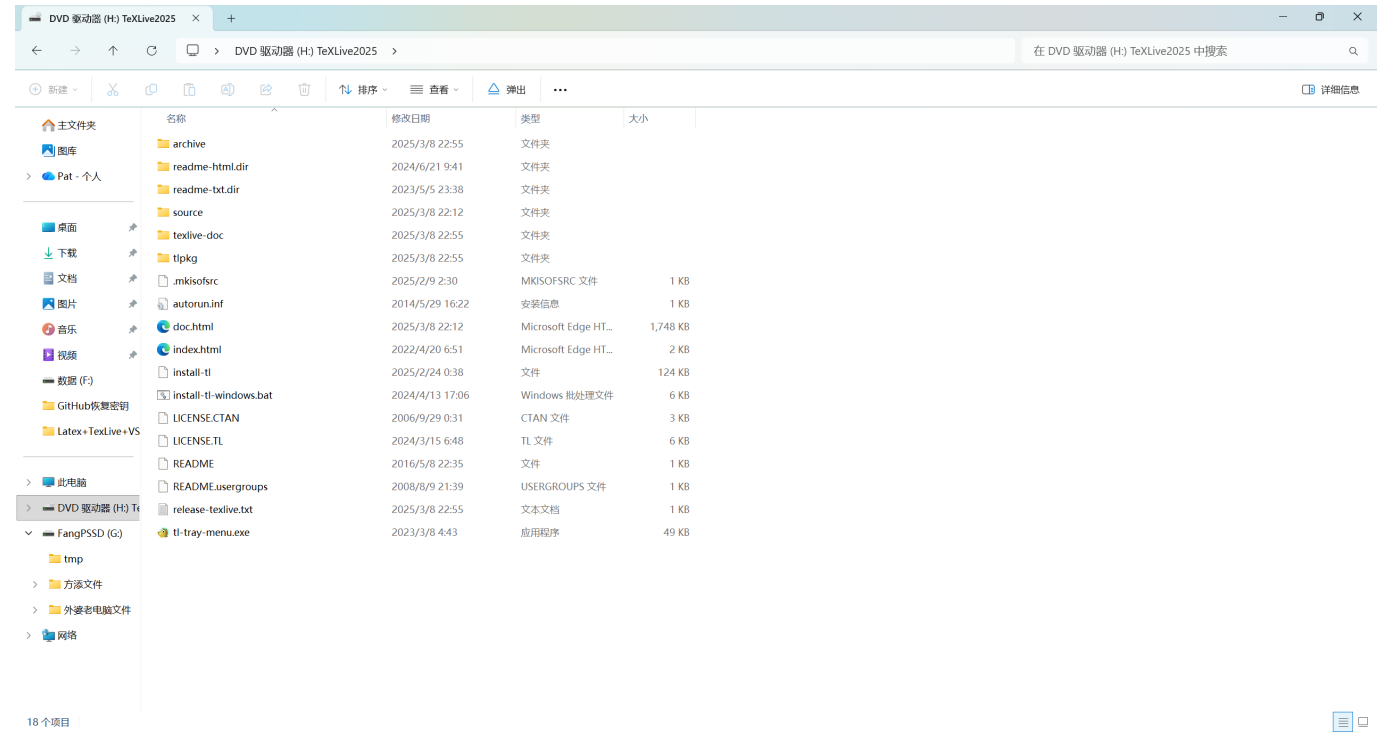
TexLive官网：<https://www.tug.org/texlive/>

TexLive下载地址：<https://mirrors.tuna.tsinghua.edu.cn/CTAN/systems/texlive/Images/>



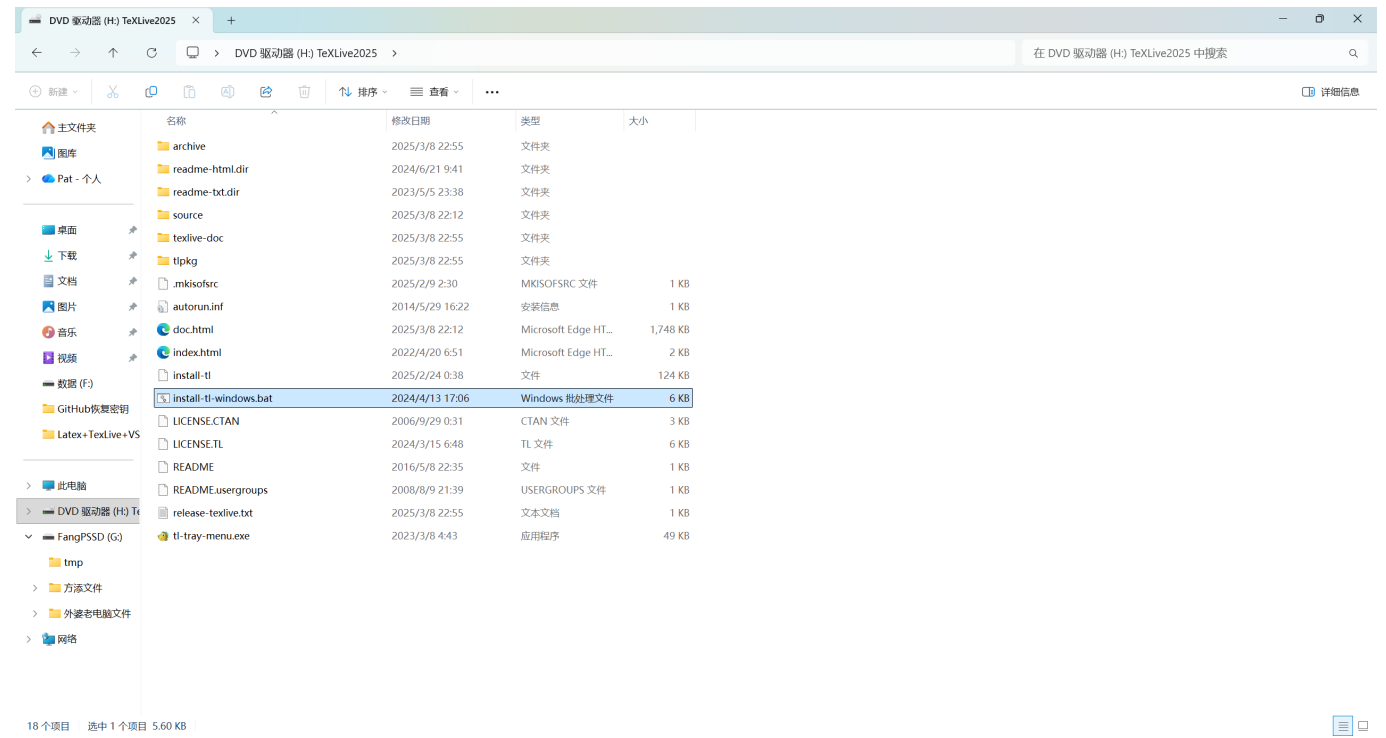
下载texlive.iso后打开镜像。



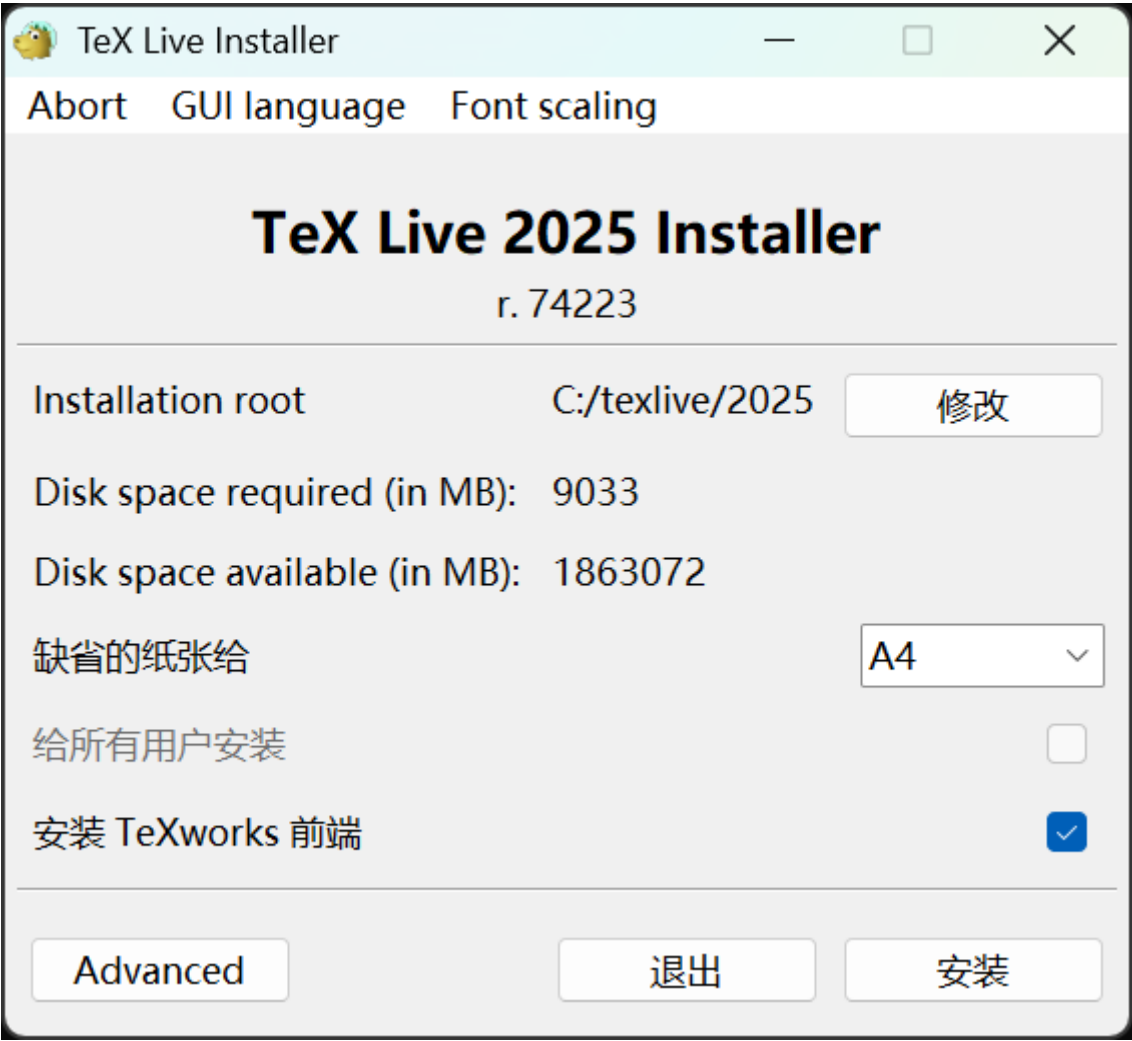


1.2 安装

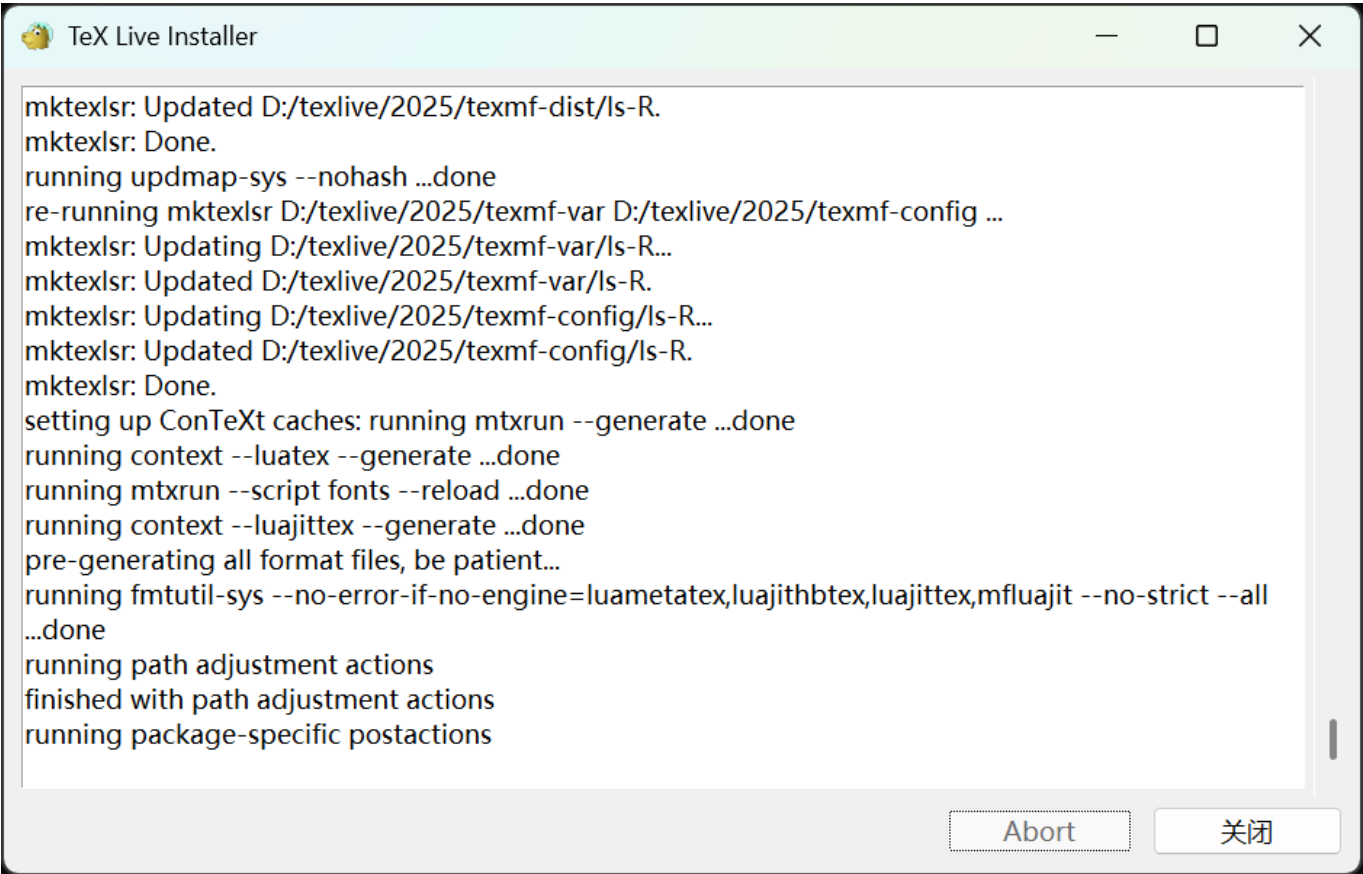
打开其中的install-tl-windows.bat



进行相应设置

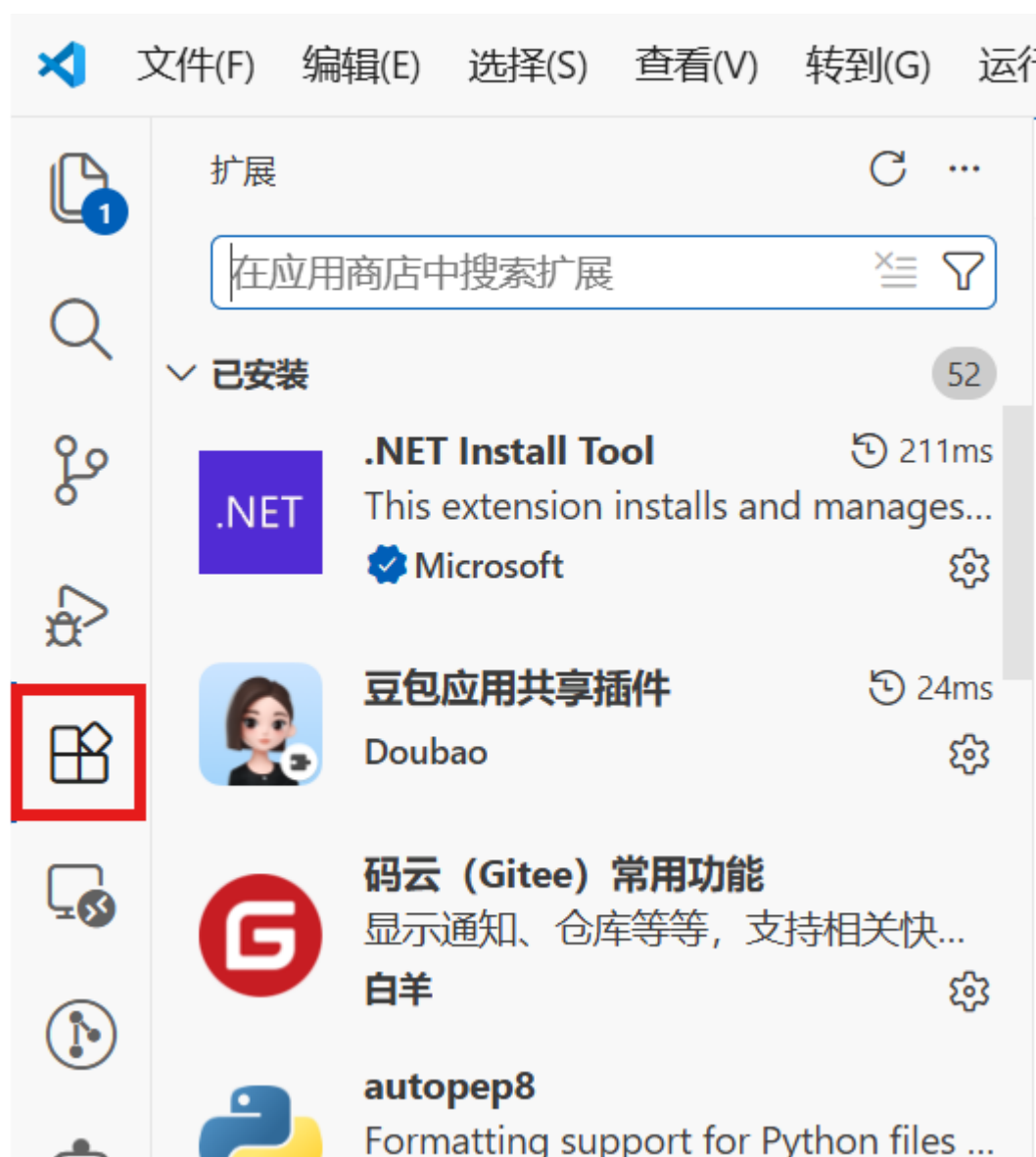


完成后点击关闭

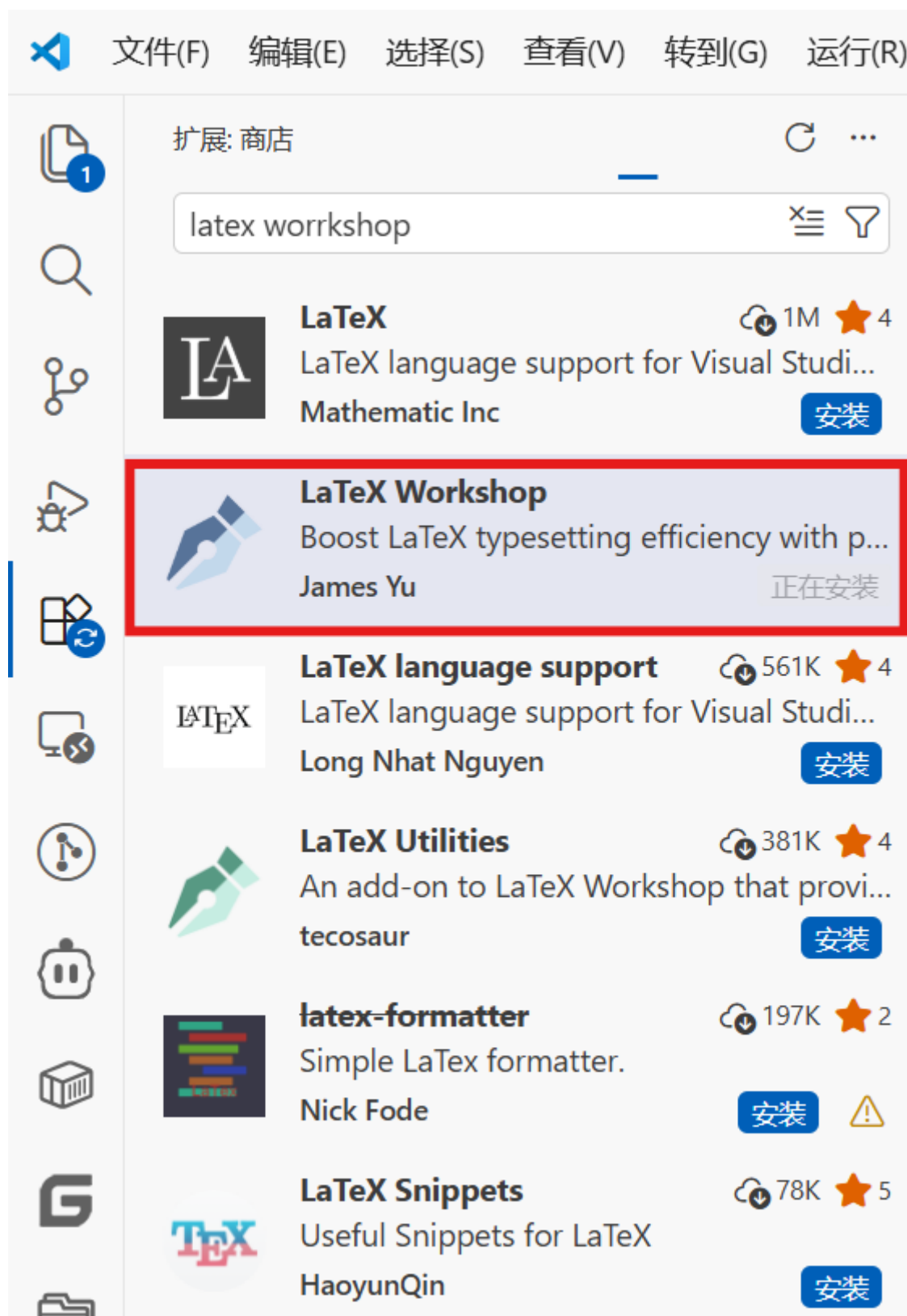


2. 安装VSCode并配置

VSCode官网: <https://code.visualstudio.com/>



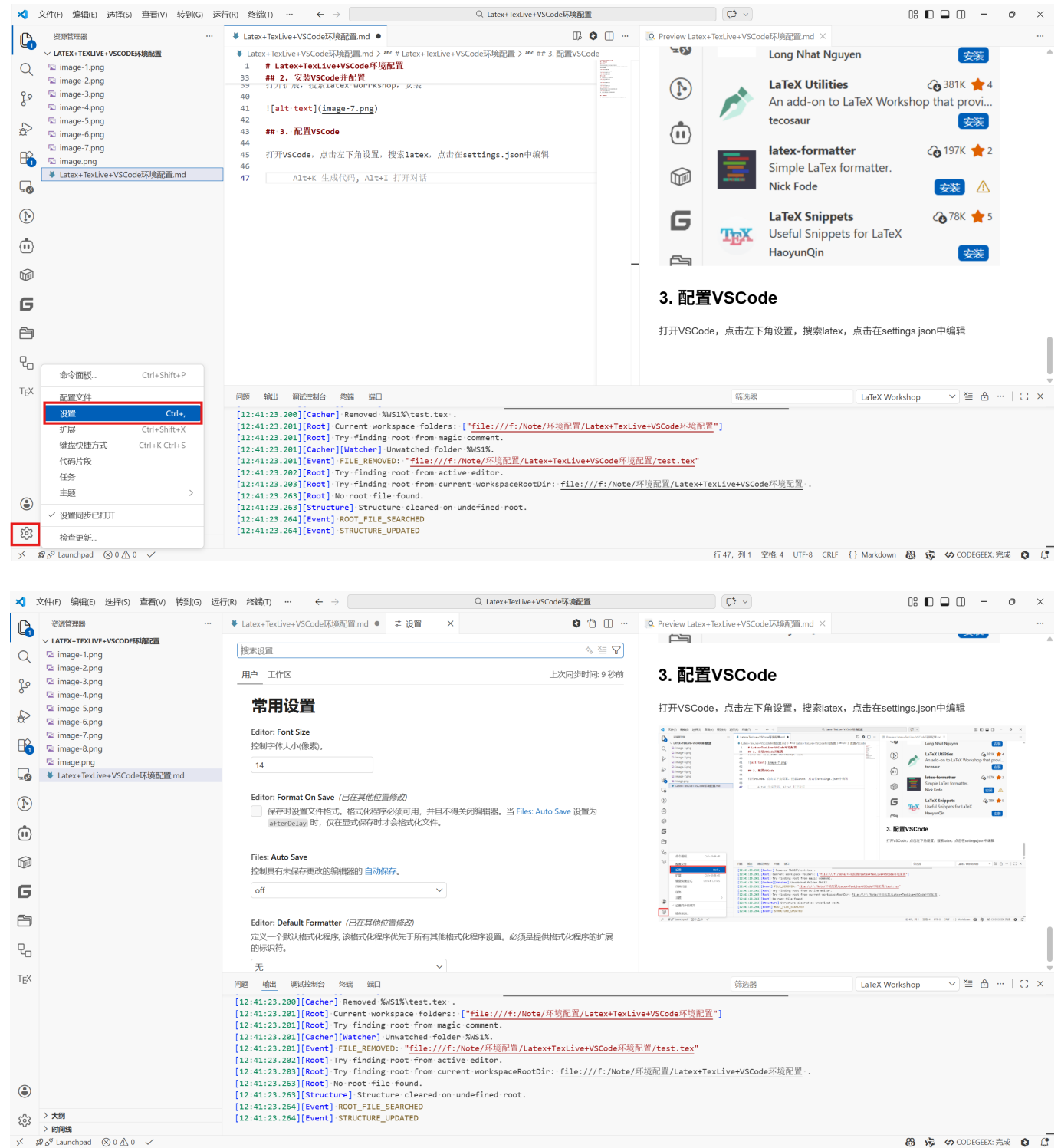
打开扩展，搜索latex workshop，安装



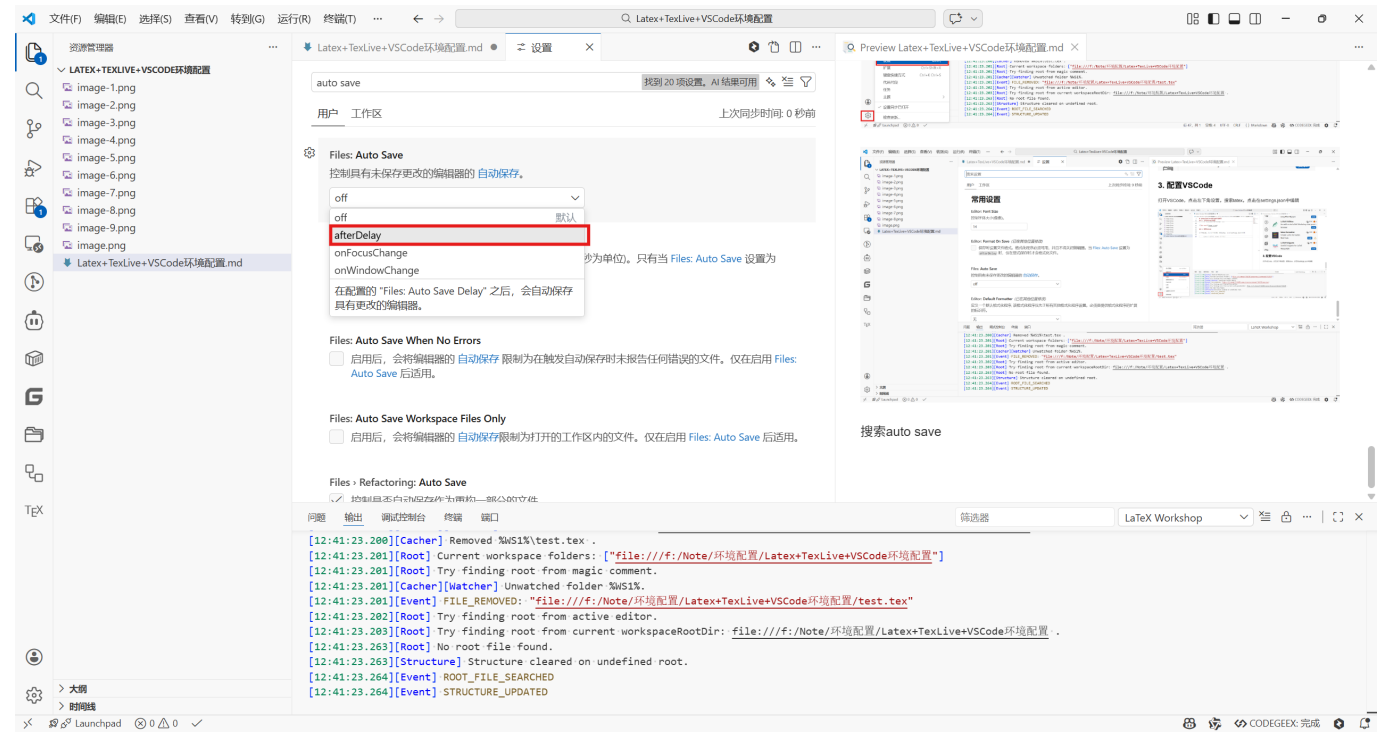
3. 配置VSCode

3.1 自动保存

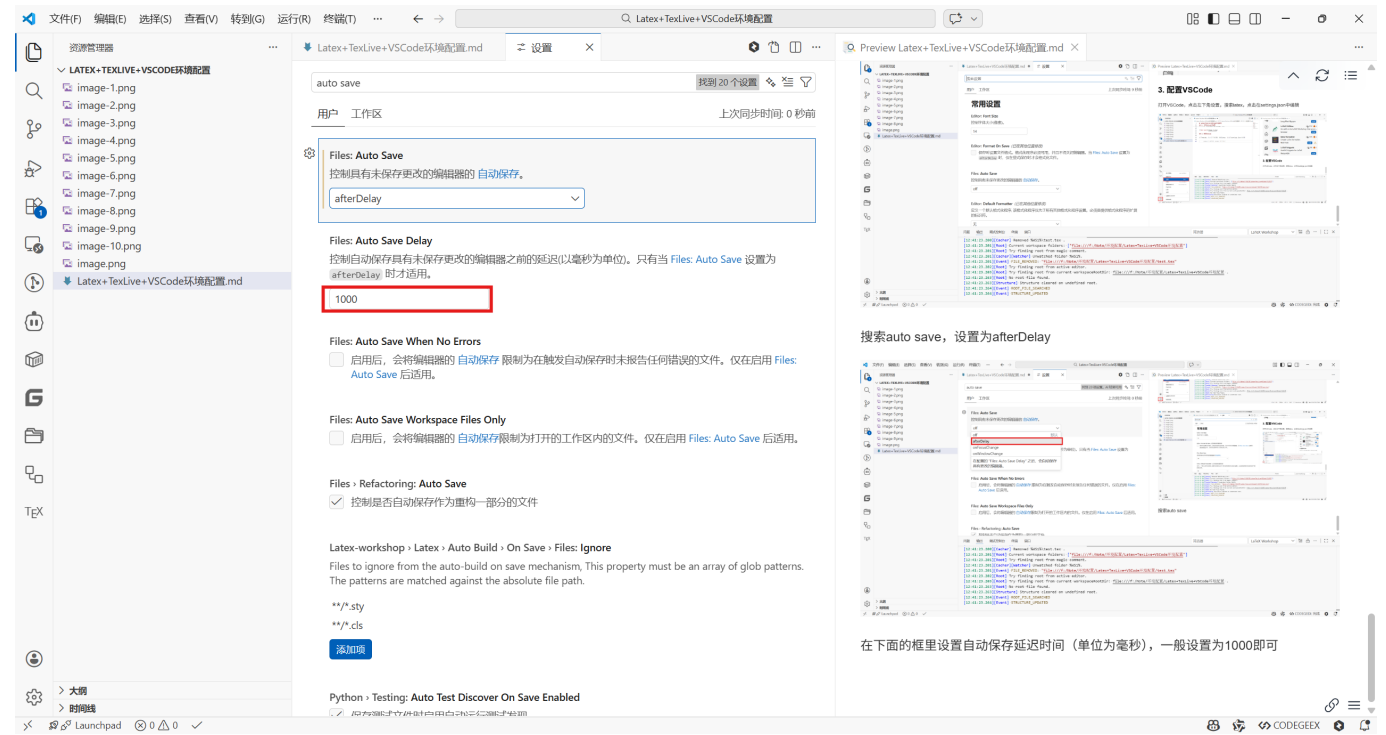
打开VSCode, 点击左下角设置



搜索auto save，设置为afterDelay

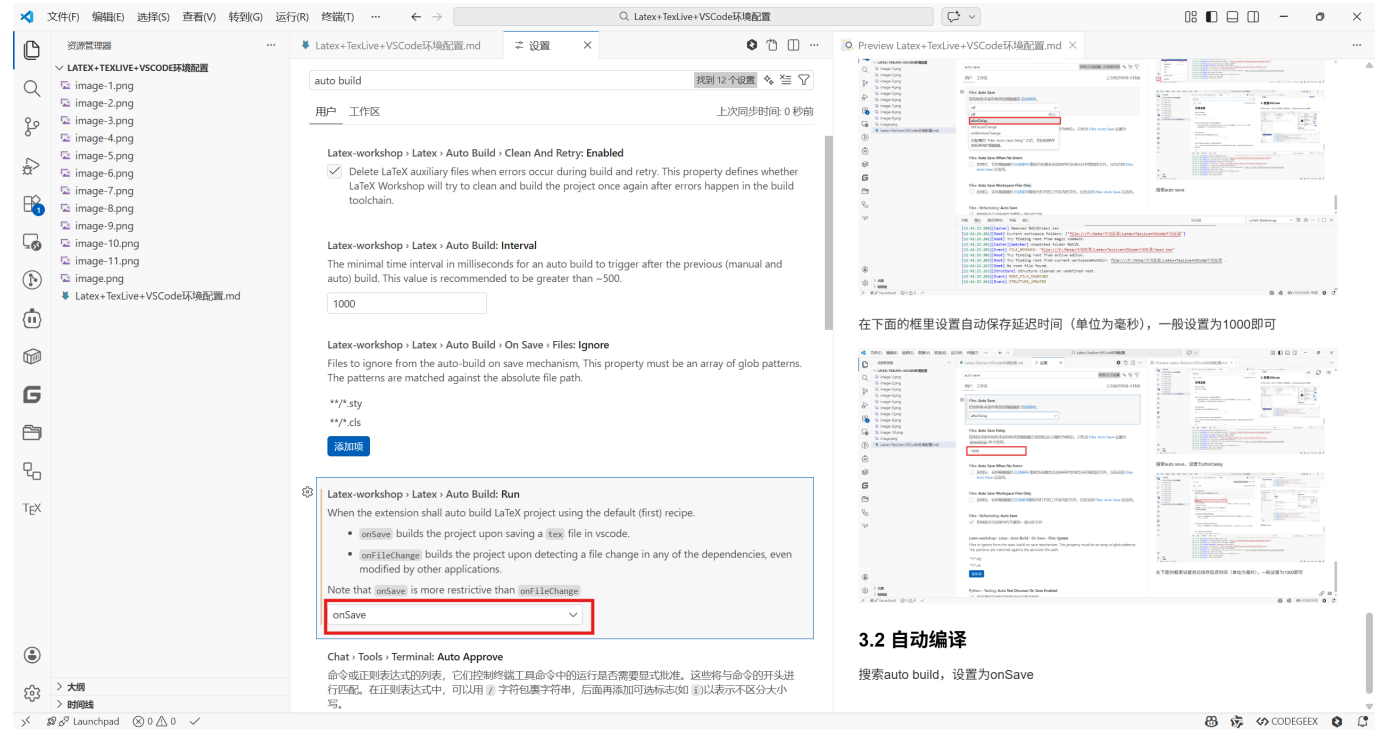


在下面的框里设置自动保存延迟时间（单位为毫秒），一般设置为1000即可



3.2 自动编译

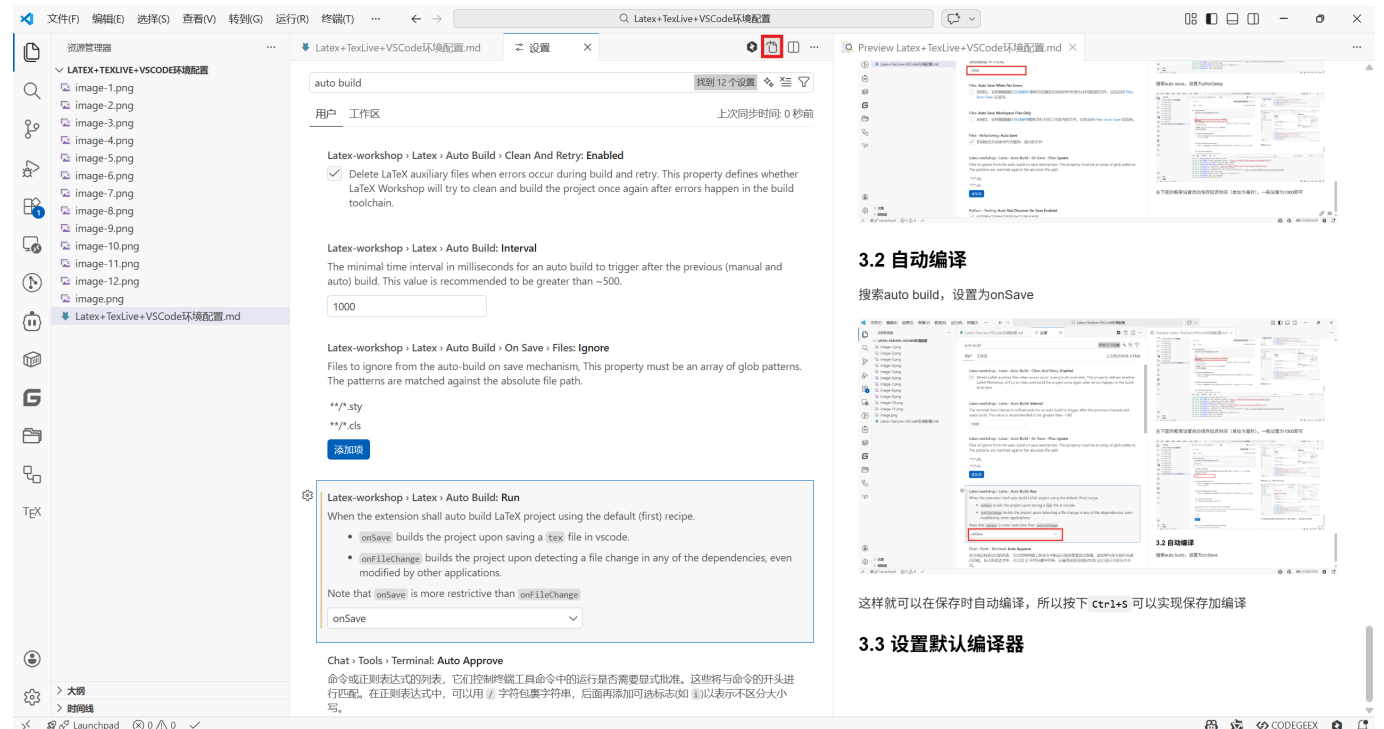
搜索auto build，设置为onSave



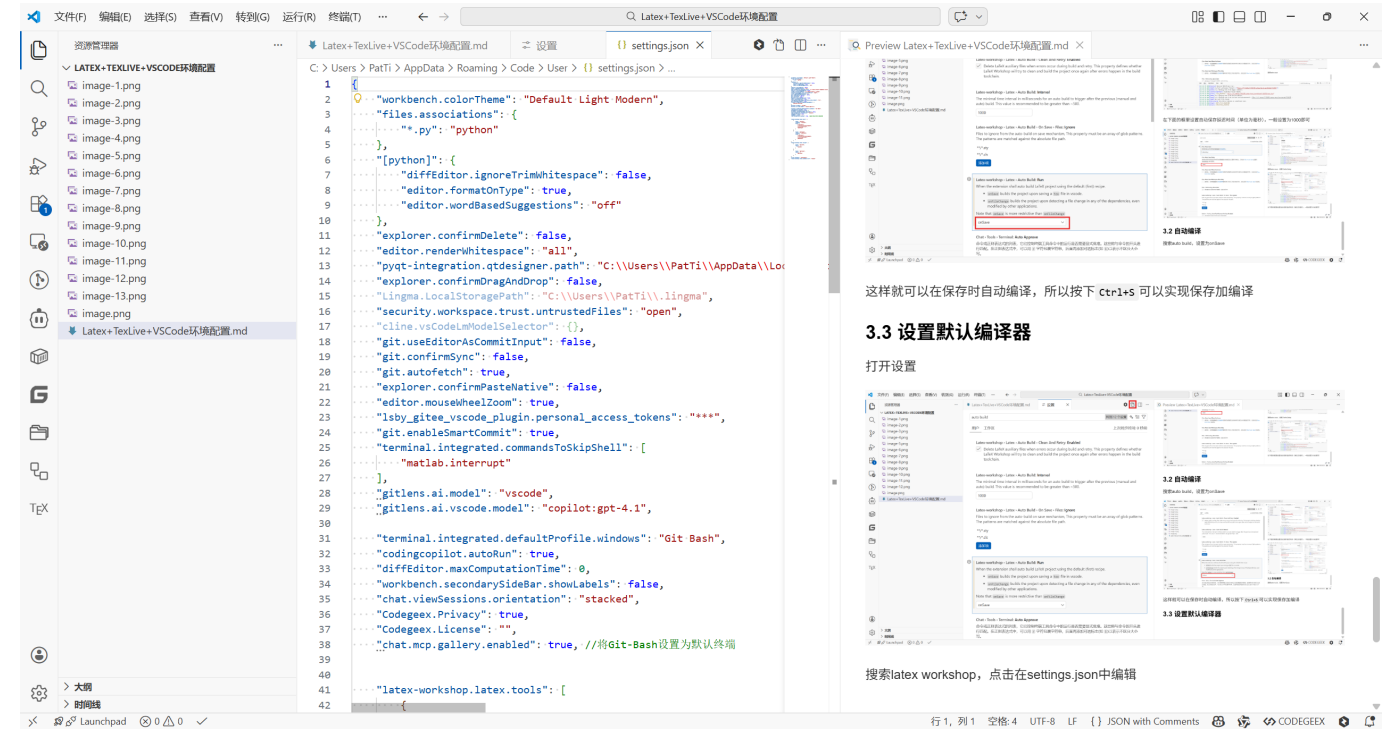
这样就可以在保存时自动编译，所以按下Ctrl+S可以实现保存加编译

3.3 设置默认编译器

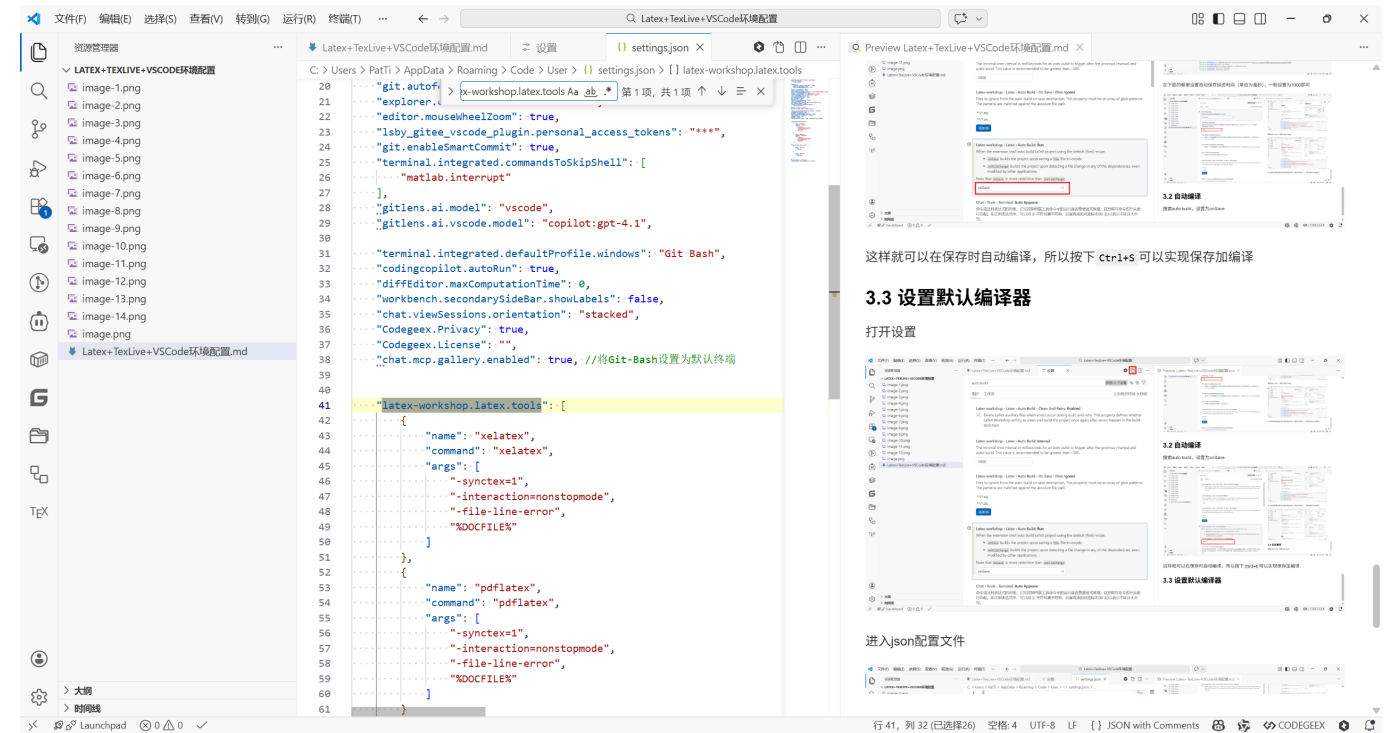
打开设置



进入json配置文件



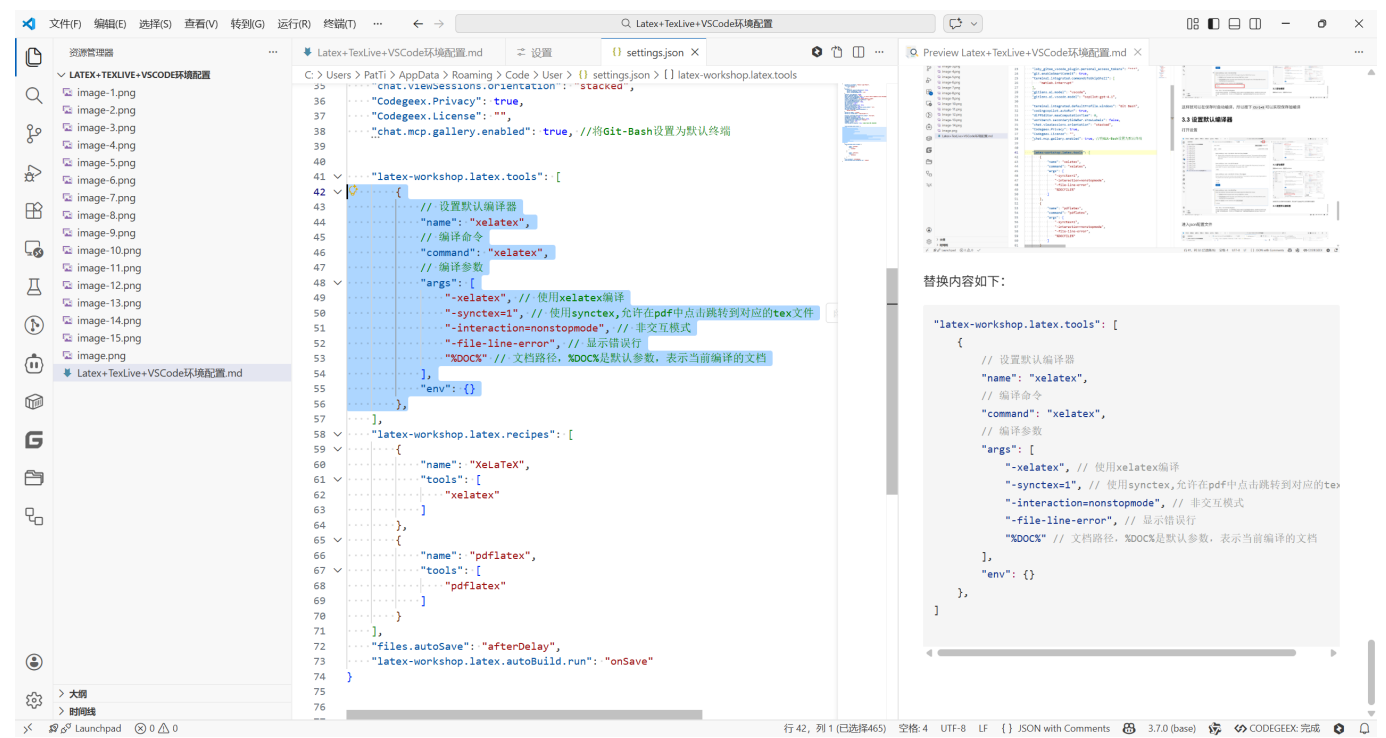
按下**Ctrl+F**搜索,latex-workshop.latex.tools



替换内容如下：

```
"latex-workshop.latex.tools": [
  {
    // 设置默认编译器
    "name": "xelatex",
    // 编译命令
    "command": "xelatex",
    // 编译参数
    "args": [
      "-synctex=1",
      "-interaction=nonstopmode",
      "-file-line-error",
      "%DOCFILES%"
    ]
  },
  {
    "name": "pdflatex",
    "command": "pdflatex",
    "args": [
      "-synctex=1",
      "-interaction=nonstopmode",
      "-file-line-error",
      "%DOCFILES%"
    ]
  },
  {
    "name": "xdocfiles",
    "command": "xdocfiles",
    "args": [
      "%DOCFILES%"
    ]
  }
]
```

```
"args": [
  "-xelatex", // 使用xelatex编译
  "-synctex=1", // 使用synctex,允许在pdf中点击跳转到对应的tex文件
  "-interaction=nonstopmode", // 非交互模式
  "-file-line-error", // 显示错误行
  "%DOC%" // 文档路径, %DOC%是默认参数, 表示当前编译的文档
],
"env": {}
},
]
```

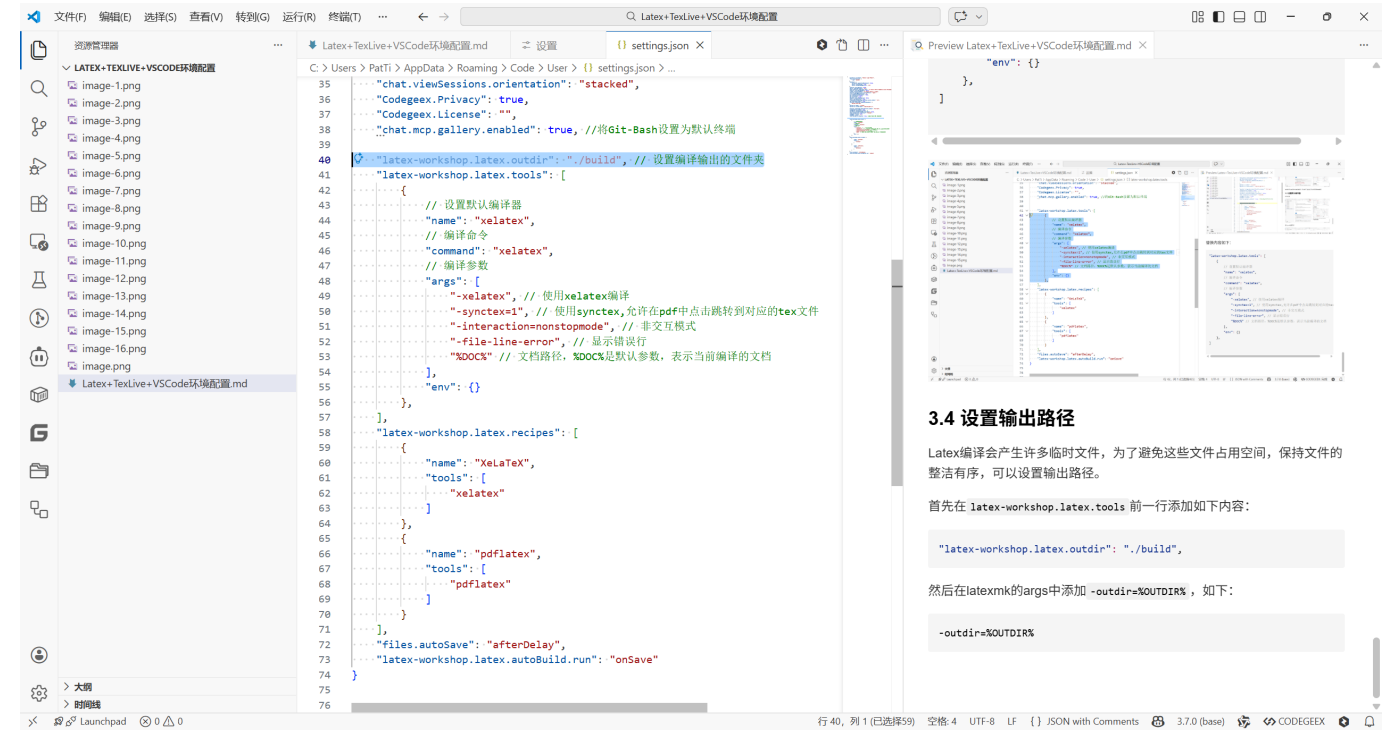


3.4 设置输出路径

Latex编译会产生许多临时文件，为了避免这些文件占用空间，保持文件的整洁有序，可以设置输出路径。

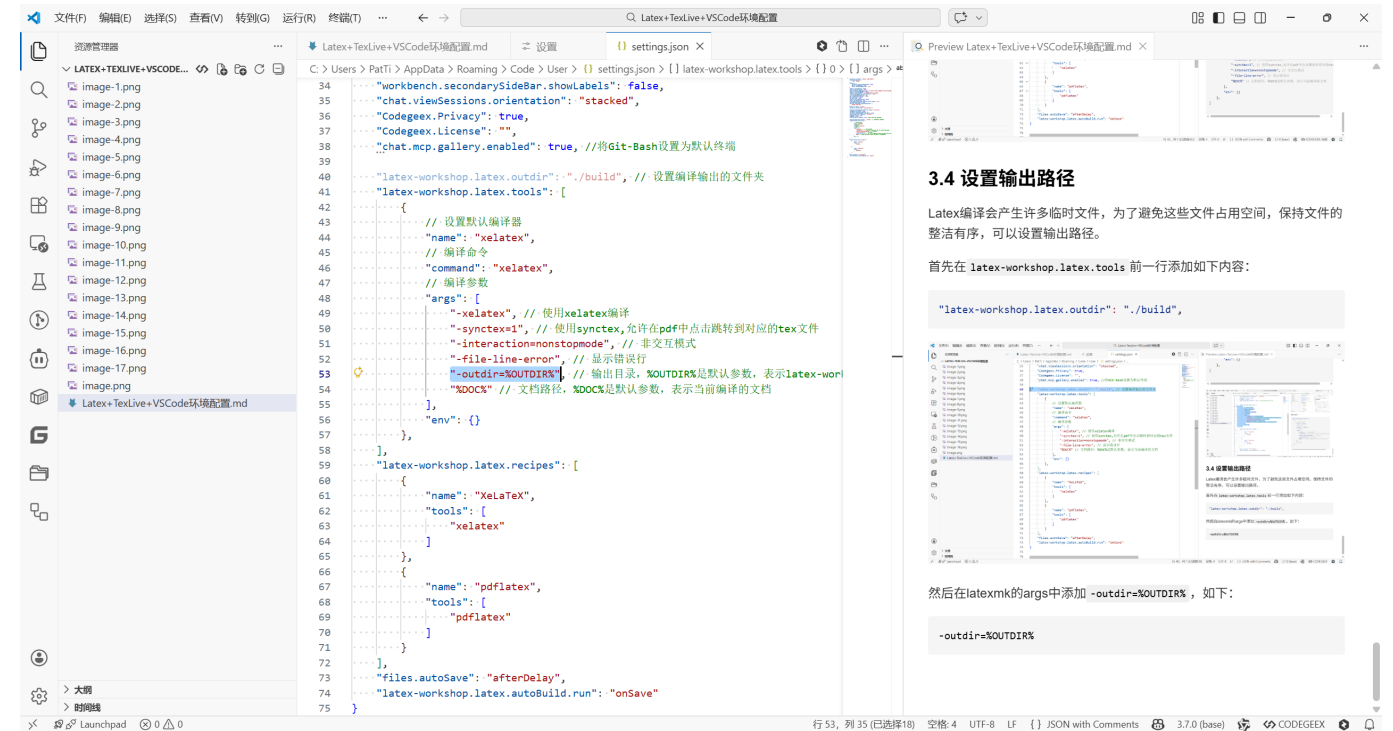
首先在`latex-workshop.latex.tools`前一行添加如下内容：

```
"latex-workshop.latex.outDir": "./build",
```



然后在latexmk的args中添加`-output-directory=%OUTDIR%`，如下：

`-output-directory=%OUTDIR%`



3.5 个人配置

笔者个人习惯PDF生成在tex同一目录下，其他编译的辅助文件（如.aux、.log、.synctex.gz等）生成在build文件夹中，目录结构如下：

```
00-Latex语法/
├── Latex语法.tex          # 源文件
├── Latex语法.pdf          # PDF文件
├── build/
│   ├── Latex语法.aux      # 辅助文件
│   ├── Latex语法.log      # 辅助文件
│   ├── Latex语法.synctex.gz
│   └── Latex语法.pdf      # 原始PDF (可忽略)
```

因为LaTeX编译器默认将所有输出文件放在同一目录,所以需要结合编译后处理来实现。

使用recipes配置, 在编译后自动将PDF复制到源目录, 最终的配置文件如下:

```
"latex-workshop.latex.outDir": "./build",
"latex-workshop.latex.tools": [
  {
    // 设置默认编译器
    "name": "xelatex",
    // 编译命令
    "command": "xelatex",
    // 编译参数
    "args": [
      "-xelatex", // 使用xelatex编译
      "-synctex=1", // 使用synctex,允许在pdf中点击跳转到对应的tex文件
      "-interaction=nonstopmode", // 非交互模式
      "-file-line-error", // 显示错误行
      "-output-directory=%OUTDIR%", // 输出目录
      "%DOC%" // 文档路径
    ],
    "env": {}
  },
  {
    "name": "pdflatex",
    "command": "pdflatex",
    "args": [
      "-synctex=1",
      "-interaction=nonstopmode",
      "-file-line-error",
      "-output-directory=%OUTDIR%",
      "%DOC%"
    ],
    "env": {}
  },
  {
    "name": "copy-pdf",
    "command": "powershell",
    "args": [
      "-Command",
      "Copy-Item",
      "-Path",
```

```

        "%DIR%/build/%DOCFILE%.pdf",
        "-Destination",
        "%DIR%/build/%DOCFILE%.pdf",
        "-Force"
    ],
    "env": {}
}
],
"latex-workshop.latex.recipes": [
    {
        "name": "xelatex + copy PDF",
        "tools": [
            "xelatex",
            "copy-pdf"
        ]
    },
    {
        "name": "XeLaTeX",
        "tools": [
            "xelatex"
        ]
    },
    {
        "name": "pdflatex",
        "tools": [
            "pdflatex"
        ]
    }
],
"files.autoSave": "afterDelay",
"latex-workshop.latex.autoBuild.run": "onSave"
}

```